

2. A circular helix is given by

$$\mathbf{x}(t) = (a \cos t, a \sin t, ct), \quad t \in \mathbb{R}.$$

Calculate the tangent $\mathbf{t}$, curvature κ , principal normal $\mathbf{n}$, binormal $\mathbf{b}$ and torsion τ . Give a sketch of the curve indicating the directions of the vectors $\{\mathbf{t}, \mathbf{n}, \mathbf{b}\}$.

3. Show that a planar curve $\mathbf{x}(t) = (x(t), y(t), 0)$ has curvature

$$\kappa(t) = \frac{|\dot{x}\ddot{y} - \dot{y}\ddot{x}|}{(\dot{x}^2 + \dot{y}^2)^{3/2}}.$$

Find the minimum and maximum values of the curvature on the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

4. Evaluate explicitly each of the line integrals

$$\int (x \, dx + y \, dy + z \, dz), \quad \int (y \, dx + x \, dy + dz), \quad \int (y \, dx - x \, dy + e^{x+y} \, dz)$$

along (i) the straight line path from the origin to $(1, 1, 1)$, and (ii) the parabolic path given parametrically by $(x, y, z) = (t, t, t^2)$ from $t = 0$ to $t = 1$. For which of these integrals do the two paths give the same results, and why?

5. Consider the vector fields $\mathbf{F}(\mathbf{x}) = (3x^2yz^2, 2x^3yz, x^3z^2)$ and $\mathbf{G}(\mathbf{x}) = (3x^2y^2z, 2x^3yz, x^3y^2)$. Compute the line integrals $\int \mathbf{F} \cdot d\mathbf{x}$ and $\int \mathbf{G} \cdot d\mathbf{x}$ along the following paths, each of which consist of straight line segments joining the specified points:

(i) $(0, 0, 0) \rightarrow (1, 1, 1),$

(ii) $(0, 0, 0) \rightarrow (0, 0, 1) \rightarrow (0, 1, 1) \rightarrow (1, 1, 1),$

(iii) $(0, 0, 0) \rightarrow (1, 0, 0) \rightarrow (1, 1, 0) \rightarrow (1, 1, 1).$

Are either of the differentials $\mathbf{F} \cdot d\mathbf{x}$ or $\mathbf{G} \cdot d\mathbf{x}$ exact?

6. A curve C is given parametrically by

$$\mathbf{x}(t) = (\cos(\sin nt) \cos t, \cos(\sin nt) \sin t, \sin(\sin nt)) \quad 0 \leq t \leq 2\pi,$$

where n is some fixed integer. Sketch the curve. Show that

$$\oint_C \mathbf{F} \cdot d\mathbf{x} = 2\pi, \quad \text{where} \quad \mathbf{F}(\mathbf{x}) = \left(-\frac{y}{x^2 + y^2}, \frac{x}{x^2 + y^2}, 0 \right)$$

and C is traversed in the direction of increasing t . Can $\mathbf{F}(\mathbf{x})$ be written as the gradient of a scalar function? Comment on your results.

[Hint: when sketching the curve, you may find it helpful to use spherical polar coordinates, defined by $x = r \sin \theta \cos \phi$, $y = r \sin \theta \sin \phi$ and $z = r \cos \theta$ with $\theta \in [0, \pi]$ and $\phi \in [0, 2\pi)$.]

7. Use the substitution $x = r \cos \phi$, $y = \frac{1}{2}r \sin \phi$, to evaluate

$$\int_D \frac{x^2}{x^2 + 4y^2} dA$$

where D is the region between the two ellipses $x^2 + 4y^2 = 1$, $x^2 + 4y^2 = 4$.

8. A closed curve C in the $z = 0$ plane consists of the arc of the parabola $y^2 = 4ax$ ($a > 0$) between the points $(a, \pm 2a)$ and the straight line joining $(a, \mp 2a)$. The area inclosed by C is D . Show, by calculating the integrals explicitly, that

$$\oint_C (x^2 y dx + xy^2 dy) = \int_D (y^2 - x^2) dA = \frac{104}{105} a^4$$

12*. Compute the volume of the region V defined by the intersection of the three cylinders

$$V = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 \leq 1, y^2 + z^2 \leq 1, z^2 + x^2 \leq 1\}.$$

2. The vector field $\mathbf{B}(\mathbf{x})$ is everywhere parallel to the normals of a family of surfaces $f(\mathbf{x}) = \text{constant}$. Show that $\mathbf{B} \cdot (\nabla \times \mathbf{B}) = 0$.

The tangent vector at each point on a curve is parallel to a non-vanishing vector field $\mathbf{H}(\mathbf{x})$. Show that the curvature of the curve is given by $\kappa = |\mathbf{H} \times (\mathbf{H} \cdot \nabla) \mathbf{H}| / |\mathbf{H}^3|$.

3. Let $\phi(\mathbf{x})$ be a scalar field and $\mathbf{v}(\mathbf{x})$ a vector field. Show, using suffix notation, that

$$\nabla \cdot (\phi \mathbf{v}) = (\nabla \phi) \cdot \mathbf{v} + \phi(\nabla \cdot \mathbf{v}), \quad \nabla \times (\phi \mathbf{v}) = (\nabla \phi) \times \mathbf{v} + \phi(\nabla \times \mathbf{v}).$$

Evaluate the divergence and curl of the following:

$$r\mathbf{x}, \quad \mathbf{a}(\mathbf{x} \cdot \mathbf{b}), \quad \mathbf{a} \times \mathbf{x}, \quad \mathbf{x}/r^3,$$

where $r = |\mathbf{x}|$ and $\mathbf{a}, \mathbf{b}$ are constant vectors.

4. For vector fields $\mathbf{u}(\mathbf{x})$ and $\mathbf{v}(\mathbf{x})$, use suffix notation to show that,

$$\nabla \times (\mathbf{u} \times \mathbf{v}) = \mathbf{u}(\nabla \cdot \mathbf{v}) + (\mathbf{v} \cdot \nabla)\mathbf{u} - \mathbf{v}(\nabla \cdot \mathbf{u}) - (\mathbf{u} \cdot \nabla)\mathbf{v},$$

Show also that

$$(\mathbf{u} \cdot \nabla)\mathbf{u} = \nabla \left(\frac{1}{2} |\mathbf{u}|^2 \right) - \mathbf{u} \times (\nabla \times \mathbf{u})$$

5. Verify directly that the vector field

$$\mathbf{u}(\mathbf{x}) = (e^x(x \cos y + \cos y - y \sin y), e^x(-x \sin y - \sin y - y \cos y), 0)$$

is *irrotational* and express it as the gradient of a scalar field ϕ . Check that $\mathbf{u}$ is *solenoidal* and show that it can be written as the curl of the vector field $\mathbf{v} = (0, 0, \psi)$, for some function ψ .

6. Check that the following vector field is irrotational

$$\mathbf{F} = (3x^2 \tan z - y^2 e^{-xy^2} \sin y, (\cos y - 2xy \sin y) e^{-xy^2}, x^3 \sec^2 z)$$

Find the most general scalar potential $\phi(\mathbf{x})$ such that $\mathbf{F} = \nabla \phi$.

7*. Suppose $\mathbf{F} : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is divergence free, i.e. $\nabla \cdot \mathbf{F} = 0$. Show that $\mathbf{F} = \nabla \times \mathbf{A}$ where

$$\mathbf{A}(\mathbf{x}) = \int_0^1 \mathbf{F}(t\mathbf{x}) \times (t\mathbf{x}) dt.$$

What goes wrong with this formula if $\mathbf{F}$ is not defined on all of $\mathbb{R}^3$?

8. Let (u, v, w) be a set of orthogonal curvilinear coordinates for $\mathbb{R}^3$. Show that

$$dV = h_u h_v h_w du dv dw.$$

Confirm that $dV = \rho d\rho d\phi dz$ and $dV = r^2 \sin \theta dr d\theta d\phi$ are cylindrical and spherical polars respectively.

9. If $\mathbf{a}$ is constant vector and $r = |\mathbf{x}|$, verify that

$$\nabla(r^n) = nr^{n-2}\mathbf{x}, \quad \nabla(\mathbf{a} \cdot \mathbf{x}) = \mathbf{a}$$

using (i) Cartesian coordinates and suffix notation, (ii) cylindrical polar coordinates, (iii) spherical polar coordinates, and (iv) a first principles, coordinate-independent approach.

[Note: for parts (ii) and (iii) you will need to be careful with the components of $\mathbf{a}$ with respect to each of the relevant bases.]

10. The vector field $\mathbf{A}(\mathbf{x})$ is, in Cartesian, cylindrical and spherical polar coordinates respectively,

$$\mathbf{A}(\mathbf{x}) = -\frac{1}{2}y\mathbf{e}_x + \frac{1}{2}x\mathbf{e}_y = \frac{1}{2}\rho\mathbf{e}_\phi = \frac{1}{2}r\sin\theta\mathbf{e}_\phi.$$

Compute the $\nabla \times \mathbf{A}$ in each different coordinate system and check that your answers agree.

11. Recall that in cylindrical polar coordinates

$$\nabla = \mathbf{e}_\rho \frac{\partial}{\partial \rho} + \mathbf{e}_\phi \frac{1}{\rho} \frac{\partial}{\partial \phi} + \mathbf{e}_z \frac{\partial}{\partial z} \quad \text{and} \quad \frac{\partial \mathbf{e}_\rho}{\partial \phi} = \mathbf{e}_\phi, \quad \frac{\partial \mathbf{e}_\phi}{\partial \phi} = -\mathbf{e}_\rho,$$

while all other derivatives are zero. Derive expressions for the $\nabla \cdot \mathbf{A}$ and $\nabla \times \mathbf{A}$ where $\mathbf{A}$ is an arbitrary vector field given in cylindrical polars by $\mathbf{A} = A_\rho\mathbf{e}_\rho + A_\phi\mathbf{e}_\phi + A_z\mathbf{e}_z$. Also derive an expression for the Laplacian of a scalar function $\nabla^2 f$ in this coordinate system

12. By applying the divergence theorem to the vector field $\mathbf{a} \times \mathbf{A}$, where $\mathbf{a}$ is an arbitrary constant vector and $\mathbf{A}(\mathbf{x})$ is a vector field, show that

$$\int_V \nabla \times \mathbf{A} \, dV = \int_S d\mathbf{S} \times \mathbf{A}$$

where $S = \partial V$. Verify this result when $V = \{(x, y, z) : 0 < x < a, 0 < y < b, 0 < z < c\}$ and $\mathbf{A}(\mathbf{x}) = (z, 0, 0)$.

13. Let $\mathbf{F}(\mathbf{x}) = (x^3 + 3y + z^2, y^3, x^2 + y^2 + 3z^2)$ and let S be the *open* surface

$$1 - z = x^2 + y^2, \quad 0 \leq z \leq 1.$$

Use the divergence theorem and cylindrical polar coordinates to evaluate $\int_S \mathbf{F} \cdot d\mathbf{S}$. Verify your result by calculating the area integral directly.

[Hint: you should find that $d\mathbf{S} = (2\rho \cos \phi, 2\rho \sin \phi, 1) \rho \, d\rho \, d\phi$.]

1. Consider the line integral

$$I = \oint_C -x^2 y \, dx + xy^2 \, dy$$

for C a closed curve traversed anti-clockwise in the (x, y) -plane.

(i) Evaluate I when C is a circle of radius R centred at the origin. Use Green's theorem to relate the results for $R = b$ and $R = a$ to an area integral over an appropriate region, and calculate the area integral directly.

(ii) Now suppose C is the boundary of a square centred at the origin with sides of length ℓ . Show that I does not change if the square is rotated in the (x, y) -plane.

2. Verify Stokes' theorem for the hemispherical shell $S = \{x^2 + y^2 + z^2 = 1, z \geq 0\}$, and the vector field

$$\mathbf{F}(\mathbf{x}) = (y, -x, z).$$

3. By applying Stokes' theorem to the vector field $\mathbf{a} \times \mathbf{F}$ for $\mathbf{a}$ constant, or otherwise, show that for a vector field $\mathbf{F}(\mathbf{x})$

$$\oint_C d\mathbf{x} \times \mathbf{F} = \int_S (d\mathbf{S} \times \nabla) \times \mathbf{F}$$

where $C = \partial S$. Verify this result when C is the boundary of a unit square lying in the (x, y) -plane, with opposite vertices at $(0, 0, 0)$ and $(1, 1, 0)$, and $\mathbf{F}(\mathbf{x}) = \mathbf{x}$.

4. Let $S = \{\mathbf{x} : |\mathbf{x}| = 1\}$ be the surface of a unit sphere. For the vector field

$$\mathbf{F}(\mathbf{x}) = \frac{\mathbf{x}}{r^3}$$

where $r = |\mathbf{x}|$, compute the integral $\int_S \mathbf{F} \cdot d\mathbf{S}$. Deduce that there *does not* exist a vector potential for $\mathbf{F}$, i.e. there can be no $\mathbf{A}$ for which $\mathbf{F} = \nabla \times \mathbf{A}$. Compute $\nabla \cdot \mathbf{F}$ and comment on your result.

5*. Consider the following vector field

$$\mathbf{A}(\mathbf{x}) = \frac{1}{(x^2 + y^2)r} (yz, -xz, 0)$$

where $r = |\mathbf{x}|$. Compute $\nabla \times \mathbf{A}$. Does this contradict the result of Question 4? Apply Stokes' theorem to $\nabla \times \mathbf{A}$ on the open surface

$$S_\epsilon = \{\mathbf{x} : |\mathbf{x}| = 1, x^2 + y^2 \geq \epsilon^2\}$$

How does this help reconcile the existence of $\mathbf{A}$ with the result of Question 4?